STAT6011

COMPUTATIONAL STATISTICS AND BAYESIAN LEARNING

Lecture 2

Gaussian Quadrature

Gaussian Quadrature

- The basic form of a Gaussian quadrature rule is

$$\int_a^b W(x)f(x)dx = \sum_{i=1}^{n} w_i f(x_i), \tag{1}$$

  where both the weights w_i and the nodes x_i are determined to give an accurate approximation.

- In Gaussian quadrature, by varying both the weights and nodes, formulas are obtained which are exact when $f(x)$ is a polynomial of degree $2n-1$.

- To compute $\int g(x)dx$, note that let $f(x) = \frac{g(x)}{W(x)}$,

$$\int g(x)dx = \int W(x)\frac{g(x)}{W(x)}dx = \int W(x)f(x)dx.$$

Hermite Polynomial

- $H_n(x)$ is a set of orthogonal polynomials over the domain $(-\infty, \infty)$ with weight function $\exp(-x^2)$.
- The first few Hermite polynomials are

$$\begin{aligned} H_0(x) &= 1, \\ H_1(x) &= 2x, \\ H_2(x) &= 4x^2 - 2, \\ H_3(x) &= 8x^3 - 12x. \end{aligned}$$

In general,

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x).$$

- Orthogonality means that

$$\int H_i(x)H_j(x)d\Phi(x) = 0$$

for $i \neq j$ and as $H_0(x) = 1$,

$$\int H_0(x)H_j(x)d\Phi(x) = \int H_j(x)d\Phi(x) = 0. \qquad (2)$$

- Equation (2) is always satisfied when x_i's are the roots of H_m (it can be shown that H_m has m distinct real roots). So, we can solve H_m to obtain $x_i, i = 1, \ldots, m$.

- To obtain the weight w_i, make (1) exact for $H_k(x)$, where $k = 0, 1, \ldots, m-1$. It is derived that

$$w_i = \frac{2^{m-1} m! \sqrt{\pi}}{m^2 [H_{m-1}(x_i)]^2}$$

- *Example*: Let $m = 3$; we solve $H_3(x) = 8x^3 - 12x$ to obtain $x_1 = 0$, $x_2 = \frac{\sqrt{6}}{2}$, and $x_3 = -\frac{\sqrt{6}}{2}$; and thus $w_1 = \frac{2\sqrt{\pi}}{3}$, $w_2 = w_3 = \frac{\sqrt{\pi}}{6}$

Weight Functions

- If we pick a different weight function $W(x)$, the polynomial will also be different.
- Some commonly used weight functions are:

Table 1: Weight functions in Gaussian quadrature rules

Name	$W(x)$	(a, b)	Option
Legendre	1	$(-1, 1)$	`Legendre`
Chebyshev 1	$1/\sqrt{1-x^2}$	$(-1, 1)$	`chebyshev1`
Chebyshev 2	$\sqrt{1-x^2}$	$(-1, 1)$	`chebyshev2`
Hermite	$\exp(-x^2)$	$(-\infty, \infty)$	`hermite`
Jacobi	$(1-x)^\alpha(1+x)^\beta$	$(-1, 1)$	`jacobi`
Laguerre	$x^\alpha e^{-x}$	$(0, \infty)$	`laguerre`

Transformation

- In general, the integral $\int_a^b f(x)dx$ over an arbitrary $[a, b]$ can be transformed into an integral over $[-1, 1]$ by using the change of variables

$$t = \frac{2x - a - b}{b - a} \Leftrightarrow x = \frac{1}{2}[(b - a)t + a + b]$$

- Then

$$\begin{aligned}\int_a^b g(x)dx &= \frac{b-a}{2}\int_{-1}^{1} g\{\frac{1}{2}[(b-a)t+a+b]\}dt \\ &= \frac{b-a}{2}\int_{-1}^{1} w(t)\frac{g\{\frac{1}{2}[(b-a)t+a+b]\}}{w(t)}dt \\ &= \frac{b-a}{2}\int_{-1}^{1} w(t)f(t)dt,\end{aligned}$$

where

$$f(t) = \frac{g\{\frac{1}{2}[(b-a)t+a+b]\}}{w(t)}.$$

Remarks

- When the limits of integration are finite, we can use the change of variables and then adopt the Legendre, Chebyshev 1, Chebyshev 2, or Jacobi quadrature.
- If the limits of integration are not $-\infty$ and ∞, then the Hermite weight function cannot be used.
- Similarly, if the limits of integration are not 0 and ∞, then the Laguerre weight function cannot be used.

- The Gauss-Hermite quadrature is a natural method for many statistical applications, especially for integrals arising in Bayesian calculations, since the Gauss–Hermite weight function is proportional to a normal density.
- In the Bayesian inference, the posterior distribution is often approximately normal, and this approximation improves as the sample size increases.